

Subsystem	Component	Qty.	Mass (kg)	Notes and Rational
Structure	Primary equipment decks/plates	2	5.24	Parametric sizing for rover avionics and GPR antenna footprints
Structure	Landing Legs (struts and footpads)	4	10.36	2.59 kg/leg sized to tolerate <12g vertical, 6g lateral touchdown loads
Structure	Stiffener/ brackets/ frames	1	1.88	~0.515 kg each; prevent deck deflection under landing loads
Structure	Fasteners and support hardware	1	1.42	integration allowance ~q.5-2% of structural mass; cryogenic-rated titanium M5/M6
Structure	Fastener contingency reserve	1	0.73	5% structural mass contingency; spare fasteners / integration allowance
Structure	Antenna mast & mounting bracket	1	2.14	GPR antenna pedestal, Al tube + SS fasteners + thermal isolators for nadir pointing
Structure	Radiation shielding vault enclosure	2	2.08	Ti-6Al-4V box cavity ~.3m x .25m x 0.22 m ; attenuates dose to <=1 krad/sol per REQ.01.02
Structure	Lead/polyethylene radiation liner	1	1.56	0.75" polyethylene + 0.125' lead facing selective shielding for flight computer
Structure	Thermal interface pads	2	0.34	Kapton + adodine interface; minimize thermal short through vault wall at 90K
Thermal	MLI Blanket attachment structure	1	1.44	Thin-gauge brackets & clips for MLI retention + radiator support
Thermal	Radiator support frame	2	1.52	Al tube frame for radiator mounting; thermal decoupling
Thermal	MLI blankets	1	6.32	Areal density 0.35 kg/m ³ x 18.05 m ² equivalent coverage (lander and appendages)
Thermal	Radiators / heat-pipe panels	2	9.58	4.79 kg/panel @ 6.0 kg/m ² , sized for 400-500 W rejection
Power	Battery packs (survival 12 h)	1	19.2	0.22-0.3 kg/kW.150 Wh/kg = 16 kg, +20% margin = 19.2 kg
Power	PDU and high current cabling	1	4.68	0.22-0.3 kg/kW delivered harness allowance (heritage EPS scaling)
Power	RTG	1	28	Main power source
Power	Heater	1	2.86	Heating unit
Comms	UHF radio and cabling	2	1.83	Heritage UHF radios 1.5-2kg; midpoint with harness
Comms	CX-band transponder	1	3.02	DSN-compatible SDST heritage mass ~3.0 kg
Comms	ANtennas (UHF helix /X-band LGA	1	0.86	Small helix and patch with short RF cabling
Avionics	Flight Computer	1	2.54	OBC board enclosure; comparable to small planetary avionics
Avionics	I/O / storage / interface boards	1	1.52	Logger and ADC's includes enclosure
Payload	MEDA-class environment suit	1	5.5	Perseverance MEDA mass ~5.5 kg; op power ~17W
Software	Flight / GNC/ FDIR	0	0	No physical mass; impact OBC/EPS
Harness and Integration	System harness / connectors/ strain relief	1	3.18	2-3% of dry mass if not fully counted elsewhere
			117.8	

Subsystem	Total Mass (kg)	% Dry Mass
Structure	25.75	21.86
Thermal	18.86	16.01
Power	54.74	46.47
Comms	5.71	4.85
Avionics/OBC	4.06	3.45
Payload	5.5	4.67
Harness and Integration	3.18	2.70
Software	0	0.00
TOTAL	117.8	100.00
Mass Cp Limit (kg)	147.25	
Allowable Mass Growth: 25% of dry mass	29.45	